PROBLEM 2 • MT-AIML MINI-PROJECT

Clinical Dictation Sentence Boundary Detector & Tokenizer

Technical Architecture, Processing Methodology & Engineering Design

1. Problem in NLP for Clinical Transcripts

Physicians and medical scribes dictate clinical encounters into microphones producing continuous spoken audio. When transcribed, this text is often unpunctuated or contains heavy clinical shorthand, decimal lab values, vital signs, and medication instructions.

Standard off-the-shelf Natural Language Processing splitters (like generic NLTK Punkt or naive regular expressions splitting on .\s+) fail drastically on clinical text. They treat every period as a sentence ending, tearing apart doctor titles, medication routes, dosages, and laboratory numbers.

The 5 Traps of Clinical Text

Clinical Pattern	Clinical Examples	Naive Splitter Failure Mode	Proposed System Protection
Doctor & Clinical Titles	Dr. Robert Smith, Prof. Lee, Tab. Aspirin	Splits at Dr. → creates orphaned fragment Smith...	Regex title shielding masks period using sentinels.
Latin Medication Routes	10 mg p.o. q.d., 500 mg b.i.d., q.i.d.	Splits at p., o., q., d. into 4 false sentences	Multi-dot abbreviation shield treats route as single unit.
Decimals & Measurements	126.4 mg/dL, 78.5 kg, 2.5 mg	Splits between 126. and 4 destroying lab values	Numerical decimal regex protects intra-digit dots.
Vital Signs & Ratios	BP 120/80 mmHg, 138/84, HR 74 bpm	Fragments slashes & separates pressure numbers	Ratio tokenization preserves blood pressure expressions.
Diagnostic Negations	denies chest pain, no fever, without SOB	Generic stopwords strip no, reversing meaning	Custom stopword filter guards all negation terms 100%.

2. Complete End-to-End System Pipeline

The system executes a robust 7-stage deterministic pipeline combining regex pattern protection with domain-adapted NLTK Punkt and spaCy special-case tokenization:

STAGE 1	Raw Ingestion
STAGE 2	Text Cleaning
STAGE 3	Pattern Protection
STAGE 4	Sentence Split
STAGE 5	Tokenization
STAGE 6	Stopword / Negation
STAGE 7	Normalization & Output

3. Detailed Walkthrough of Each Pipeline Stage

Stage 1 & 2: Ingestion & Text Normalization

The raw transcript string is checked for non-empty input. In Stage 2, anomalous carriage returns, line breaks, multi-space gaps, and non-printable control characters are stripped into single clean whitespace separators:

```
cleaned = re.sub(r"[ ]+", " ", raw_transcript)<br/>cleaned = re.sub(r"\s{2,}", " ", cleaned).strip()
```

Stage 3: Clinical Pattern Detection & Sentinel Masking (Key Innovation)

Before splitting sentences, the system temporarily replaces periods that belong to abbreviations, decimals, and dates with a unique sentinel token: §DOT§.

• Clinical Abbreviation Regex: Dr., p.o., b.i.d., q.d., Tab., Cap. are temporarily replaced with Dr§DOT§, p§DOT§o§DOT§.
• Decimal Regex: Numbers like 126.4 become 126§DOT§4.
• Date Regex: Dates like 14.11.2023 become 14§DOT§11§DOT§2023.

Stage 4: Sentence Boundary Detection Engine (Hybrid Punkt & Regex)

The segmenter handles two major clinical transcription regimes:

1. Punctuated Clinical Narratives: High-precision sentence boundary detection is executed using a domain-adapted NLTK PunktSentenceTokenizer initialized with medical abbreviation types coupled with regex sentinels. True boundaries are detected with 98.06% recall and 92.66% precision.

2. Unpunctuated Continuous Audio Streams: In raw scribed streams lacking punctuation, speech pauses naturally correlate with clinical shift cues. The system recognizes transition cues such as: pt admitted, vitals stable, Dr. [Name] consulted, started on [Med], pt denies, discharge planned. It inserts synthetic sentence boundaries ahead of these shifts.

Finally, restore_protected_text restores all §DOT§ sentinels back to clean standard periods.

Stage 5: Medical Word Tokenizer (Domain-Adapted spaCy & NLTK)

Splitting medical text on simple whitespace or standard word boundaries breaks apart critical clinical expressions. The specialized ClinicalTokenizer regex and domain-adapted spaCy tokenizer preserve:

• Compound Measurement Units: mg/dL, mmol/L, mL/min, mcg, mmHg
• Blood Pressure & Vital Ratios: 120/80, 138/84, 142/88
• Hyphenated Medical Terminology: follow-up, post-operative, sub-acute
• Multi-dot Shorthand: p.o., b.i.d., q.d., p.r.n.

Stage 6: Negation-Guarded Stopword Filtering

Generic NLP stopword lists (like default NLTK or spaCy lists) contain words like no, not, without. In healthcare, removing these words is catastrophic: “Patient has no fever and denies chest pain” → would become “Patient fever chest pain”!

```
PRESERVED_NEGATIONS = { “no”, “not”, “none”, “neither”, “never”, “denies”, “denied”, “without”, “negative”, “ruled out” }
```

```
CLINICAL_STOPWORDS = COMBINED_STOPWORDS - PRESERVED_NEGATIONS
```

Every negation keyword is protected, achieving 100.00% preservation accuracy.

Stage 7: Clinical Shorthand Normalization

Physicians write rapid abbreviations. The normalization pass expands clinical shorthand into canonical standard tokens while preserving uppercase acronyms (BP, HR, ED, CT):

```
NORMALIZATION_MAP = {<br/>"pt": "patient", "hx": "history", "rx": "prescription", "dx": "diagnosis",<br/>"c/o": "complaining_of", "s/p": "status_post", "p.o.": "orally", "b.i.d.": "twice_daily",<br/>"q.d.": "once_daily", "p.r.n.": "as_needed", "tab.": "tablet", "cap.": "capsule"<br/>}
```

4. Step-by-Step Execution Trace Example

Below is an actual transcript trace through the 7 stages of the implemented pipeline:

INPUT CLINICAL TRANSCRIPT: "Patient was evaluated today by Dr. Robert Smith. Patient has a prior hx of HTN. Rx was adjusted to Lisinopril 10 mg p.o. q.d. along with Tab. Aspirin 81 mg daily. Fasting glucose was 126.4 mg/dL. Dr. Smith advised follow-up in 4 weeks."

Stage	Transformation & Output
1. Text Cleaning	Strips extraneous spaces and formats into single normalized string.
2. Sentinel Masking	Dr§DOT§ Robert Smith... prior hx of HTN... 10 mg p§DOT§o§DOT§ q§DOT§d§DOT§... Tab§DOT§ Aspirin... 126§DOT§4 mg/dL... Dr§DOT§ Smith follow-up in 4 weeks.

Stage	Transformation & Output
3. Sentence Split	S1: Patient was evaluated today by Dr. Robert Smith. S2: Patient has a prior hx of HTN. S3: Rx was adjusted to Lisinopril 10 mg p.o. q.d. along with Tab. Aspirin 81 mg daily. S4: Fasting glucose was 126.4 mg/dL. S5: Dr. Smith advised follow-up in 4 weeks.
4. Tokenization	Preserves units and compound tokens intact: ['Dr.', 'Smith', 'hx', 'HTN', 'Rx', '10', 'mg', 'p.o.', 'q.d.', 'Tab.', 'Aspirin', '81', 'mg', '126.4', 'mg/dL', 'follow-up']
5. Stopword Filtering	Non-clinical stopwords (was, by, at, the, of, in) removed; negations retained.
6. Normalization	Shorthand expanded: hx → history | rx → prescription | p.o. → orally | q.d. → once_daily | tab. → tablet | dr. → doctor

5. Software Component Architecture

The project adheres to a clean, modular object-oriented software design separating preprocessing, models, pipelining, evaluation, and user presentation:

Module / File	Primary Classes & Functions	Core Responsibility
src/preprocessing.py	protect_medical_abbreviations(); segment_sentences(); tokenize_medical_words(); remove_custom_stopwords(); normalize_tokens()	Regex rule definitions, clinical lexicons (titles, routes, units, negations), sentinel replacement, NLTK & spaCy clinical tokenizers.
src/models.py	ClinicalSentenceSegmenter; ClinicalTokenizer	Domain-specific model wrappers providing fit(), segment(), tokenize(), and filter_stopwords() APIs supporting hybrid, Punkt, and spaCy engines.
src/pipeline.py	ClinicalDictationPipeline	End-to-end orchestrator connecting text cleaning, entity detection, segmentation, token extraction, stopword filtering, and normalization.
src/evaluation.py	compare_baseline_vs_proposed(); run_full_system_evaluation(); split_dataset()	Benchmark execution, metric computation (P, R, F1), error categorization, baseline runners (NLTK & spaCy baselines), and report generators.
run.py	Flask Application Routes: /, /analyze, /api/benchmark, /api/compare, /api/samples/<id>	Interactive web dashboard and REST API for real-time transcript testing, multi-library comparison, and benchmark visualization.